Quiz 06. Euler-Cauchy equations
prepared by C. Cahig *for* ENS181 (AY 2026-2027, S1)

Instructions. In the following problems, the character d is reserved for derivatives and differentials so that $\mathrm{d}\alpha$ is "the differential of α". As is customary, e is the base of the natural logarithm, and $\exp(abc) = e^{abc}$.

First find the general solutions of following Euler-Cauchy equations. Use the method of variation of parameters.

Equation 1.

$$4(3-2x)^3y''' + 32(3-2x)^2y'' + 128(3-2x)y' + 256y = -2(-2x+3)^4y^{(4)}$$

Equation 2.

$$162\left(1-\frac{2}{3}t\right)^4 w^{(4)} + 864\left(1-\frac{2}{3}t\right)^2 w'' + 128w = 648\left(1-\frac{2}{3}t\right)^3 w''' + 288\left(1-\frac{2}{3}t\right)w'$$

Equation 3.

$$\frac{\mathrm{d}^3F}{\mathrm{d}r^3} - \frac{3}{r}\frac{\mathrm{d}^2F}{\mathrm{d}r^2} + \frac{6}{r^2}\frac{\mathrm{d}F}{\mathrm{d}r} = \frac{3}{r^3} + \frac{3\ln r}{r^3}$$

Scenario 1. Find the particular solutions $y(x)$, $w(t)$, and $F(r)$ corresponding to the following initial conditions.

- *for Equation 1:*

$$y(1) = 1, \quad y'(1) = -2, \quad y''(1) = 8, \quad y'''(1) = -48$$

- *for Equation 2:*

$$w(1) = 0, \quad w'(1) = -4, \quad w''(1) = 4, \quad w'''(1) = 92$$

- *for Equation 3:*

$$F(1) = 2, \quad F'(1) = \tfrac{414}{121}, \quad F''(1) = \tfrac{466}{121}$$

Compute the following values.

1. $y(-6.9000)$
2. $y(-6.2275)$
3. $y(-5.5550)$
4. $y(-4.8825)$
5. $y(-4.2100)$
6. $w(-6.9000)$
7. $w(-6.2275)$
8. $w(-5.5550)$
9. $w(-4.8825)$
10. $w(-4.2100)$
11. $F(6.9000)$
12. $F(6.2275)$
13. $F(5.5550)$
14. $F(4.8825)$
15. $F(4.2100)$

Scenario 2. Find the particular solutions $y(x)$, $w(t)$, and $F(r)$ corresponding to the following boundary conditions.

- *for Equation 1:*

$$y(1) = 1, \quad y'(1) = -2, \quad y(1.5-0.5e) = e + e^2, \quad y''(1.5-0.5e) = -2 - 8e$$

- *for Equation 2:*

$$w(1) = 0, \quad w'(1) = -4, \quad w(1.5-0.5e^{\pi}) = -2, \quad w'''(1.5-0.5e^{\pi}) = -2e^{\pi}$$

- *for Equation 3:*

$$F(1) = 2, \quad F'(1) = \frac{414}{121}, \quad F(2) = 1 + 8\cos\left(\sqrt{2}\ln 2\right) + \frac{3}{22}\ln^2 2 + \frac{51}{121}\ln 2$$

Compute the following values.

16. $y(-6.9000)$
17. $y(-6.2275)$
18. $y(-5.5550)$
19. $y(-4.8825)$
20. $y(-4.2100)$
21. $w(-6.9000)$
22. $w(-6.2275)$
23. $w(-5.5550)$
24. $w(-4.8825)$
25. $w(-4.2100)$
26. $F(6.9000)$
27. $F(6.2275)$
28. $F(5.5550)$
29. $F(4.8825)$
30. $F(4.2100)$

Marking. Each required item above is worth two (2) points. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.